

Arjun Mauji

mauji.arjun@gmail.com | (647) 225-3264 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Mechanical Engineering undergraduate at Queen's University with hands-on experience in mechanical and embedded systems design, along with strong problem-solving and communication skills developed through collaborative engineering projects.

Education

B.A.Sc. Mechanical Engineering, Queen's University

Kingston, Canada | 3rd/4

- Dean's Scholar 24/25, Biggs – Ronald & Deanna Scholarship
- Relevant Coursework/Concepts: Embedded Programming, Applied Control Systems, Machine Tool Laboratory, Manufacturing Methods

Work Experience

Teaching Assistant (Mechatronics Course), Queen's University

Sep – Dec 2025

- Co-supervised 10 mechatronics labs with over 100 students on measurement sensors and signal noise filtering
- Answered electronic measurement theory questions and troubleshoot lab circuits

Projects

Oscilloscope, Individual Design Project

June 2026 – Present

- Building an STM32-based oscilloscope to sample and visualize analog signals between 0-3.3 V
- Designing an analog front-end circuit to safely condition and interface real-world signals with the onboard ADC
- Implementing a custom SPI driver in bare-metal C to interface with a display module and visualize sampled data
- Writing a custom linker script and boot process to build and flash the firmware without an IDE

Embedded Software Team, Queen's Off-Road Car Design Team (Baja)

Feb – Mar 2026

- Programmed STM32 boards in C to monitor the vehicle transmission system's thermal condition
- Assisted in migrating the vehicle's existing CAN bus communication library to use FreeRTOS

Hardware Motion Analyzer, Individual Design Project

Nov – Dec 2025

- Prototyped a low-cost (<\$40) wireless motion analyzer to compare dynamic movement flexibility between users
- Programmed 2 IMU-based sensor modules to capture movements by calculating joint flexion and extension angles
- Implemented RTOS tasking with I2C communication to log and filter up to 10 angle data streams concurrently
- Analyzed movement flexibility and peak angle values using SciPy

Mechanical Team, Queen's Off-Road Car Design Team (Baja)

Sep – Dec 2024

- Assisted in the remodelling and manufacturing of the continuously variable transmission (CVT) using SolidWorks

Skills

Mechanical: SolidWorks CAD & FEA, Onshape, Reading Assembly Drawings, PID

Embedded Systems: STM32, ESP32, FreeRTOS, CAN, I2C, SPI, UART

Software Tools: C, Python, MATLAB (Simulink), Git/GitHub

Electrical Tools: Multimeters, Soldering

Professional Tools: MS Outlook, Word, Excel, PowerPoint

Spoken Languages: Bilingual (Fluent) – English & French